

Question 2 Clustering: Average Trait Model

This notebook evaluates the custom CSC 466 random forest using `data/processed`
`/breed_clustering_avg.csv`.

This notebook clusters dog breeds using the individual AKC trait ratings with grouped average features:
family life, physical, social, and personality.

Overall Metrics
Overall silhouette score: 0.19
Rand Index: 0.76
Average radius / average intercluster distance: 0.92

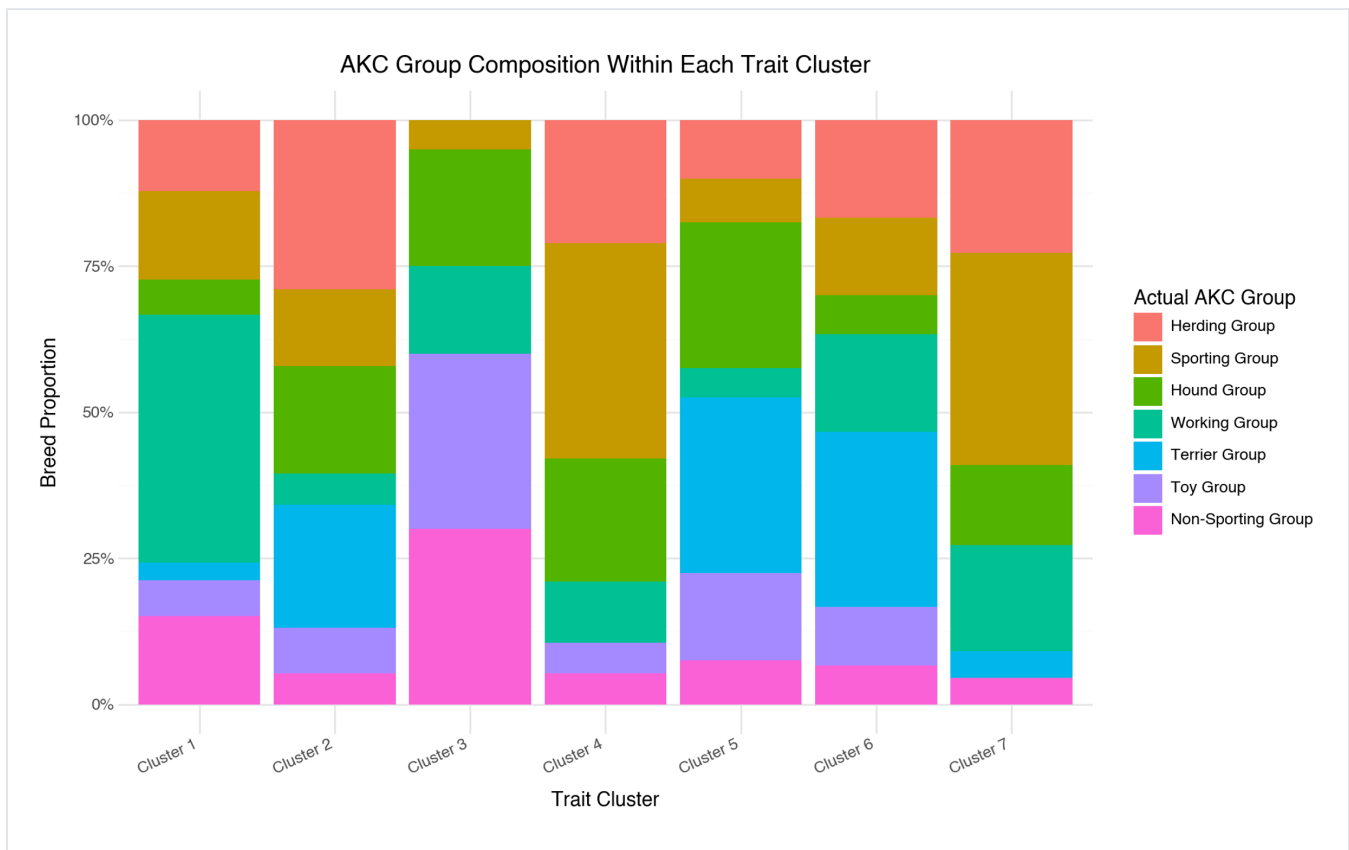
Trait Cluster	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cl
Metric					
Cluster Size	33	38	20	19	
Most Common AKC Group	Working Group	Herding Group	Non-Sporting Group	Sporting Group	Terrier
Most Common AKC Group Count	14	11	6	7	
Predicted AKC Group	Working Group	Terrier Group	Non-Sporting Group	Sporting Group	Hound
Distance To Predicted AKC Group	0.6	0.36	0.7	0.84	
AKC Rand Index	0.7	0.63	0.76	0.69	
Silhouette Score	0.17	0.17	0.11	0.24	
Radius	1.38	1.08	1.29	1.04	

Cluster Centroids vs AKC Group Traits Averages

	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5	Cluster 6	Cluster 7
Trait							
Family Life	3.34	3.82	4.67	4.89	3.42	4.21	4.45
Physical	2.70	1.87	2.27	2.21	1.72	2.33	2.74
Social	3.46	3.59	3.80	3.95	3.29	4.32	3.48
Personality	3.41	3.88	2.98	4.21	3.11	4.22	3.56

AKC Group	Herding Group	Hound Group	Non-Sporting Group	Sporting Group	Terrier Group	Toy Group	Working Group
Trait							
Family Life	3.75	4.08	4.08	4.22	3.86	4.03	3.85
Physical	2.26	1.97	2.22	2.38	1.94	2.00	2.61
Social	3.59	3.30	3.82	3.67	3.77	3.85	3.75
Personality	3.92	3.48	3.35	3.82	3.57	3.39	3.52

How well do trait-based clusters line up with AKC Groups?



Prediction Output

This table saves each breed 's assigned cluster with its average rank, popularity tier, and AKC group.

	Breed	Trait Cluster	Predicted AKC Group	Actual AKC Group	Distance To Predicted AKC Group	Correct AKC Prediction
0	Affenpinscher	Cluster 1	Working Group	Toy Group	0.60	False
1	Afghan Hound	Cluster 5	Hound Group	Hound Group	0.80	True
2	Airedale Terrier	Cluster 5	Hound Group	Terrier Group	0.80	False
3	Akita	Cluster 1	Working Group	Working Group	0.60	True
4	Alaskan Malamute	Cluster 1	Working Group	Working Group	0.60	True
...	...	...	...	...	...	...
197	Wire Fox Terrier	Cluster 6	Sporting Group	Terrier Group	0.76	False
198	Wirehaired Pointing Griffon	Cluster 6	Sporting Group	Sporting Group	0.76	True
199	Wirehaired Vizsla	Cluster 6	Sporting Group	Sporting Group	0.76	True
200	Xoloitzcuintli	Cluster 2	Terrier Group	Non-Sporting Group	0.36	False
201	Yorkshire Terrier	Cluster 6	Sporting Group	Toy Group	0.76	False

202 rows × 6 columns